


Castor bean tick *Ixodes ricinus*

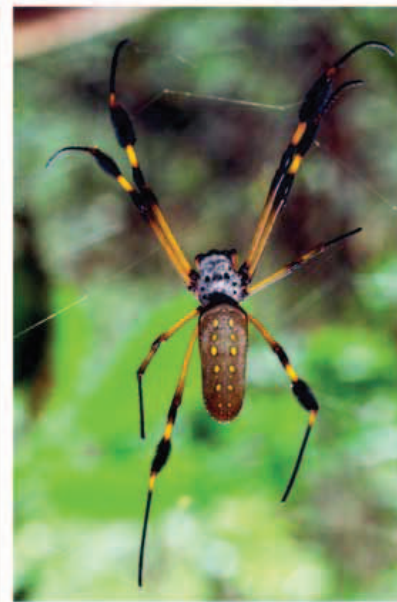
Ticks and mites

Order Acari

Species 48,200

Occurrence Most terrestrial habitats

Many of the tiny, eight-legged mites and ticks are harmless, but some are parasites that latch onto warm-blooded animals—including humans—and suck their blood. A feeding tick can consume so much blood that it swells to up to 10 times its normal size before dropping off to breed. In the process, it may transmit infections such as Siberian tick typhus—a disease that can kill if not treated. Other, much smaller mites may cause itching diseases such as scabies.


Golden Orb *Nephila* species

Spiders

Order Araneae

Species About 42,000

Occurrence Most terrestrial habitats

The most familiar arachnids, spiders are eight-legged predators that either hunt down their prey or snare it in complex traps made of silk. They kill their victims with venom injected by a pair of sharp fangs, then flood the tissues with digestive juices to liquefy them—as, unlike insects, spiders cannot eat solid food. Normally only used for hunting, the venom of some species, such as the black widows, includes nerve toxins that can kill a human.

Mollusks

Phylum Mollusca

Bivalves

Class Bivalvia

Species 15,000

Occurrence Fresh- and seawater habitats

Mollusks have soft bodies that are protected by hard shells. A bivalve, such as a clam, has two shells hinged together so that they can be sealed shut to enclose the animal within. Most bivalves live attached to rocks or buried in soft sediments.

Gastropods

Class Gastropoda

Species 75,000

Occurrence Aquatic habitats; wet, damp, and dry regions on land

Gastropods such as snails have a single coiled shell into which they retreat to escape danger or stop themselves from drying out. They glide on a layer of mucus secreted by the flat “foot” beneath the body. Land snails and slugs eat plant material, but many marine snails are predators. Tropical cone shells have a powerful venom that they use to kill prey.

Cephalopods

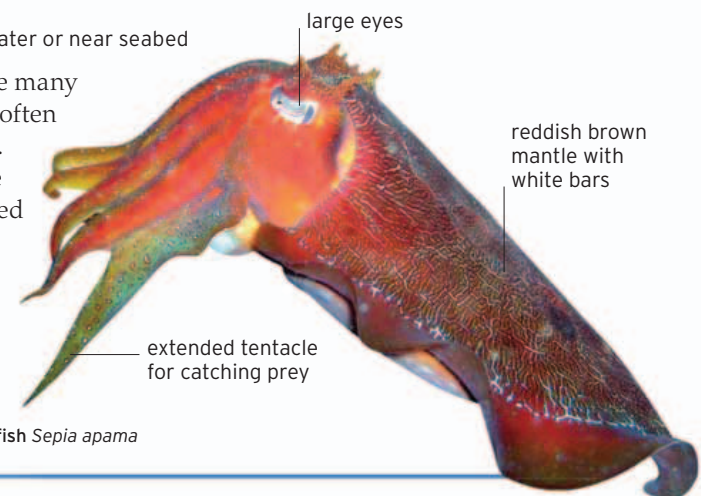
Class Cephalopoda

Species About 750

Occurrence Open water or near seabed

Cephalopods have many flexible arms and often a pair of tentacles. A few, such as the tropical blue-ringed octopus, have a highly venomous bite. Squid can swim at high speed by jetting

water from their bodies, and many species can change color instantly.


Australian giant cuttlefish *Sepia apama*

Echinoderms

Phylum Echinodermata

Starfish

Class Asteroidea

Species 1,853

Occurrence On the seabed

These spiny-skinned invertebrates have body segments that radiate like the spokes of a wheel from a central hub containing the mouth and main internal organs. This is most obvious in starfish, which typically have five long arms, although some have more than 40. They creep over the seabed on hundreds of small, hydraulic tube feet and prey on other invertebrates.

Sea urchins

Class Echinoidea

Species 1,090

Occurrence On or beneath seabed

A sea urchin has the same radial symmetry as starfish, but its body is roughly spherical and the radial elements are arranged like the segments of an orange. It has a shell-like skeleton under its skin, which supports sharp, protective spines. Most urchins eat seaweed and encrusting animals, rasping them from rocks.

Invertebrate chordates

Phylum Chordata

Subphyla Urochordata, Cephalochordata

Sea squirts

Class Ascidacea

Species 2,813

Occurrence Attached to coastal rocks and the seabed

Adult sea squirts have a spherical or cylindrical protective tunic, or test, that surrounds their bodies. They live attached to rocks and filter seawater for food—drawing water into their bodies, passing it through a sievelike

structure, and then pumping it out again. However, when young they are completely different—free-swimming animals that resemble tadpoles. Remarkably, the tails of these larvae are strengthened by a rod called a notochord, which is a feature found in vertebrates such as fish. For this reason, the simple sea squirts are classified among the chordates—a phylum that includes all vertebrates, including humans.